Grade 4 Science – Midterm Sheet (October)

Science · Mohamed Khair · Knowledge City Schools · Name: _____ Date: _____

Q1: Choose the correct answer

1. **People who invent and design products to help us are called:**
   - A Doctors
   - B Engineers
   - C Teachers

2. **Engineering is the process of using science and math to find solutions to:**
   - A Games
   - B Problems
   - C Stories

3. **Which engineer oversees the construction of roads, bridges, and skyscrapers?**
   - A Civil engineer
   - B Medical engineer
   - C Mechanical engineer

4. **Medical engineers develop devices to help:**
   - A Build roads
   - B Doctors diagnose and treat diseases
   - C Design cars

5. **Mechanical engineers study materials, heat, and:**
   - A Plants
   - B Energy
   - C Animals

6. **What do medical engineers use to perform surgery?**
   - A Lasers and robots
   - B Hammers and nails
   - C Telescopes

7. **Scientists use a microscope to see things that are:**
   - A Very large
   - B Far away in space
   - C Too small to see with bare eyes

8. **Which tool helps scientists take pictures of planets and stars?**
   - A Microscope
   - B Hubble Telescope
   - C Clock

9. **Which scientist discovered that electricity causes quartz crystal to vibrate?**
   - A Pierre Curie
   - B Doug Engelbart
   - C Bill English

10. Engineers designed watches using quartz crystals to keep:

A Temperature
B Time
C Distance

11. An electronic whiteboard in class is an example of technology in:

A Space
B Learning
C Medicine

12. A satellite orbits around a planet or a:

A Star
B Bridge
C Building

13. What is the Moon?

A An artificial satellite
B A natural satellite
C A robot

14. Satellites help scientists collect data about:

A Weather and land masses
B Bridges and buildings
C Human heart

15. What system uses satellites to help in navigation?

A X-ray
B GPS
C Laser

16. What do engineers use to launch satellites into space?

A Rockets
B Cars
C Trains

17. To "design" means to:

A Buy products
B Make or draw plans to build something
C Destroy tools

18. What is the first step in the design process?

A Build a prototype
B Test the prototype
C Identify a problem

19. After identifying a problem, engineers do:

A Research
B Redesign
C Final test

20. Who invented the computer mouse?

A Pierre Curie
B Doug Engelbart
C Warren Morrison

21. The first version of a test model built by engineers is called a:

A Graph
B Prototype
C Table

22. The first computer mouse was made of:

A Plastic
B Wood
C Glass

23. Why do engineers test a prototype?

A To see if it meets the needs
B To sell it immediately
C To throw it away

24. To share test results, engineers use tables and:

A Toys
B Graphs
C Stories

25. Communicating results helps engineers find solutions to:

A Math homework
B Problems with a design
C Daily meals

26. When engineers evaluate user reports, they may need to:

A Stop working
B Redesign the prototype
C Hide the design

27. Engineers choose the best solution that costs less money and is:

A Dangerous
B Safe to use
C Very heavy

28. Which tool is used to measure the length of a small object like a mouse?

A Ruler
B Thermometer
C Scale

29. Whiteboards enable teachers to show photos, maps, and:

A Food
B Videos
C Clothes

30. **Engineering is the process of using science and math to solve:**

A Problems
B Games
C Letters

Q2: Complete the sentences

31. **_____ engineers oversee the construction of highways and bridges.**

A natural
B Civil
C prototype
D mouse

32. **_____ engineers design X-rays, lasers, and robots for doctors.**

A natural
B Hubble Telescope
C Medical
D Civil

33. **The _____ helps scientists see objects that are too small to see with bare eyes.**

A microscope
B prototype
C Medical
D mouse

34. **The _____ helps scientists take accurate pictures of space.**

A natural
B Hubble Telescope
C mouse
D Civil

35. **The Moon is a _____ satellite orbiting Earth.**

A natural
B graphs
C Hubble Telescope
D prototype

36. **A _____ is the first working version of a solution built by engineers.**

A mouse
B Hubble Telescope
C prototype
D Medical

37. **Engineers collect data and represent it using tables and _____.**

A Hubble Telescope
B graphs
C microscope
D Medical

38. **Doug Engelbart invented the computer _____.**

A graphs
B mouse
C Medical
D prototype

39. **Engineers use science and math to find solutions to _____.**

A graphs
B problems
C Hubble Telescope
D prototype

Q3: Match

40. Match: GPS

A Uses satellites for navigation
B Helps scientists see small things
C First test model of a design
D Takes photos of stars and galaxies

41. Match: Microscope

A Helps scientists see small things
B First test model of a design
C Takes photos of stars and galaxies
D Designs machines and tools

42. Match: Prototype

A First test model of a design
B Takes photos of stars and galaxies
C Designs machines and tools
D Helps scientists see small things

43. Match: Mechanical Engineer

A Takes photos of stars and galaxies
B Uses satellites for navigation
C Helps scientists see small things
D Designs machines and tools

44. Match: Hubble Telescope

A First test model of a design
B Takes photos of stars and galaxies
C Designs machines and tools
D Uses satellites for navigation

Q4: True or False

45. Civil engineers oversee the construction of highways, bridges, and skyscrapers.

A True ✓
B False ×

46. The Moon is an artificial satellite created by engineers.

A True ✓
B False ×

47. Medical engineers use lasers and robots to help doctors perform surgery.

A True ✓
B False ×

48. Scientists use a telescope to see things that are too small to see with bare eyes.

A True ✓
B False ×

49. Doug Engelbart invented the computer mouse.

A True ✓
B False ×

50. A prototype is the final product sold in stores.

A True ✓
B False ×

51. **Engineers share their test results using tables and graphs.**

A True ✓ B False ×

Q5: Short answers

52. **Which two tools help scientists in learning or research?**

A Microscope and Hubble Telescope B Hammer and nails

C Toys and food

53. **Why do engineers share their test results with others?**

A To find problems with the design, get feedback, and improve it so the final product is safe and works well

B To sell the prototype immediately

C To hide the design from other engineers